

Agent Use of Delegation and Interaction Traceability (AUDIT)

Side Meeting, IETF-126, July 20, 2026, Vienna

Overview

Why we are here?

Discuss the interest and scope in potential work in the IETF on auditing agent workflows

- Short problem statement (5 mins)
- Proposed Audit architecture (10 mins)
- Discussion on use cases and technical requirements (10 mins)
- Potential charter scope and deliverables (10 mins)
- Open discussion (60 mins or more)
- Next steps

Problem statement

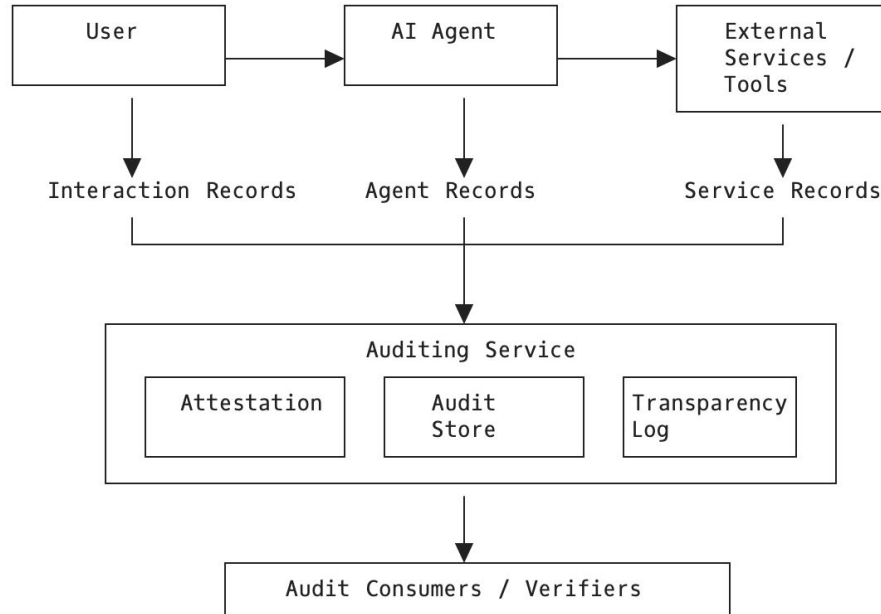
Autonomous and semi-autonomous software agents are increasingly acting on behalf of users across multiple services and administrative domains. The increased use of complex agent communications introduces significant challenges for auditability and accountability. Existing mechanisms, such as system logs, tracing systems, and authorization frameworks, capture individual aspects of system behavior but lack interoperable support for correlating user intent, delegation chains, authorization state, and resulting actions across domains. This creates challenges for fundamental questions required for compliance, operational debugging, and user trust, such as who initiated an action, under which authority it occurred, and how permissions evolved during execution.

The AUDIT effort aims to define ***interoperable protocol mechanisms and data models to enable auditing of complex, distributed, and time-evolving systems***. This includes an architectural concept, common models for audit records, propagation of audit context across interactions, and integration with existing IETF protocols such as OAuth, HTTP, and attestation and transparency frameworks.

AUDIT Architecture

[draft-kuehlewind-audit-architecture-00](#)

See also: <https://github.com/mirjak/draft-audit-architecture>

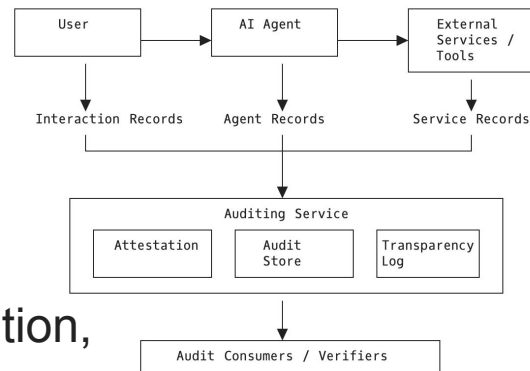


Basic Principles

- Auditing must ***correlate transitions across interactions and domains*** by maintaining common context
 - This is to enable linking user intent to resulting system actions across protocol and administrative boundaries
 - Therefore two distinct classes of interaction that must each be auditable: user-facing interactions (prompts, approvals, human-in-the-loop conformations) and system-facing interactions (API calls, tool invocations, delegation to other agents or services).
- Auditing must capture ***authorization as a time-evolving state***
 - This is to address complex agent systems that introduce dynamic, fine-grained authorization changes that arise during execution and are driven by agent decisions, sub-agent delegation, and human interaction.
 - Therefore different types of audit records must be supported that are generated distributively.

Auditing Services

- Must be logically distinct from the User, Agent, or Tool
- Stores observable records (Interaction, Action, Delegation, Authorization Transition) and provides
 - Signing with a key bound to the recorded identity
 - Registration with a Transparency Log before they leave the operational environment



Questions on use cases and requirements

- *What is this audit logging for? (What specific kinds of questions should it be able to answer?)*
- *What must be provable to someone else, rather than only recorded?*
- *What kind of confidentiality and information-leakage considerations need to be accounted for?*

Question on interoperability

- *Which contexts might it be most helpful for this work to focus on?*
- *Are we more concerned about auditability within an organization/domain, between them, or both?*
- *Are we more concerned about auditability for interactions on the open internet (e.g. between individual users and web services), or between enterprises?*

Questions on transparency logging

- *Is there a need for third-party transparency services, or is it enough for each party to have non-repudiable evidence?*
- *Does a transparency log create side-channels?*
- *Supply-chain use cases register per release; this would register per tool call, which is (many) orders of magnitude more traffic. What's the smallest thing that has to be independently registered?*

Proposed Charter (1/4) - Challenges

See also:

<https://github.com/mirjak/audit-bof-preparation/blob/main/audit-charter.md>

Autonomous and semi-autonomous software agents, including those based on LLM-based machine learning systems, are increasingly deployed to act on behalf of users, organizations, and services across the Internet. These agents interact across multiple administrative or trust domains and can initiate actions without direct human oversight at each step.

This introduces challenges for auditability, accountability, and transparency, including:

- Difficulty attributing actions to a specific user, agent instance, or delegation context
- Loss of visibility across long-running or distributed workflows
- Inconsistent capture of delegation relationships, authorization context, and identity transitions
- Cross-domain interactions lack interoperable means to exchange or verify audit-relevant information about the participating agents and their interactions

Proposed Charter (2/4) - Requirements

Agents participate in two distinct classes of interactions that must be audited:

- User-facing interactions, such as prompts, conversations, and approvals, capturing user intent and human-in-the-loop decisions
- System-facing interactions, such as API calls, tool usage, and delegation to other agents or services

Effective auditing requires linking user intent to resulting system actions across protocol and administrative boundaries. While traditional workflows support evolving authorization, these transitions are usually explicit and predefined. Agent systems introduce dynamic, fine-grained authorization changes that arise during execution, driven by agent decisions, delegation, and human interaction. Auditing must therefore capture authorization as a time-evolving state and correlate these transitions across interactions and domains.

Additionally, Agent behavior may be non-deterministic and not fully predefined, requiring auditing mechanisms to capture execution context and structure as they emerge. Auditing must also distinguish between user, agent, and service identities, and ensure audit data remains interpretable across systems without shared assumptions.

Proposed Charter (3/4) - Scope and Goals

The AUDIT working group will define interoperable mechanisms for auditing and accountability of Agents and delegated systems across Internet protocols.

The group will focus on architectures, protocol-layer specifications, and data representations that enable systems to record, exchange, and verify audit-relevant information across user-facing and system-facing interactions. This includes capturing delegation chains, evolving authorization state, and enabling consistent interpretation and correlation of audit data across domains.

The working group will compose existing IETF building blocks for identity (WIMSE), attestation (RATS), authorization (OAuth family), transparency (SCITT), context propagation (W3C Trace Context), and conversation containers (vCon), and will define only the additional protocol elements, data models, and best practices needed to make these compose coherently for the AI agent case.

The working group will not define auditing policies or compliance frameworks, but instead provide the technical building blocks needed to support them.

Proposed Charter (4/4) - Deliverables

The AUDIT working group is expected to produce:

1. **Architecture for AI Agent Auditing** An Informational RFC describing roles, trust relationships, and data flows for interoperable auditing, including the relationship between user-facing and system-facing audit signals.
2. **Audit Data Models and Semantics** One or more Standards Track RFCs defining data models for representing audit information, including interaction records, agent identity, delegation context, authorization state over time, and action provenance.
3. **Protocol Extensions or Profiles** One or more Standards Track RFCs specifying extensions to existing IETF protocols (e.g., HTTP, OAuth, or token formats) to convey audit-related information.
4. **Best Practices for Deployment and Operation** An Informational or BCP document providing guidance for secure, interoperable, and privacy-aware auditing, including correlation across interaction types.

Main questions*

- Is there interest/a need for this work?
 - What are the different use cases and requirements on logging records, attestation, and transparency?
 - Is the scope and relation to other work reasonable clear?
- Is the IETF the right place to do the work?

*This mostly covers the first BoF question (RFC5434): there is a problem that needs solving, and the IETF is the right group to attempt solving it?